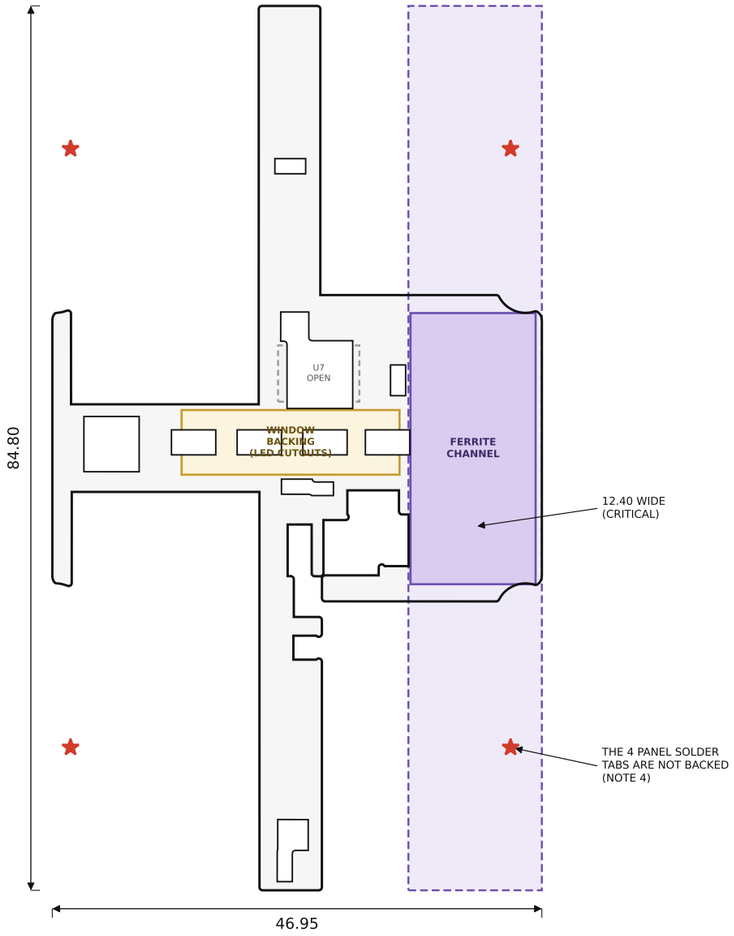
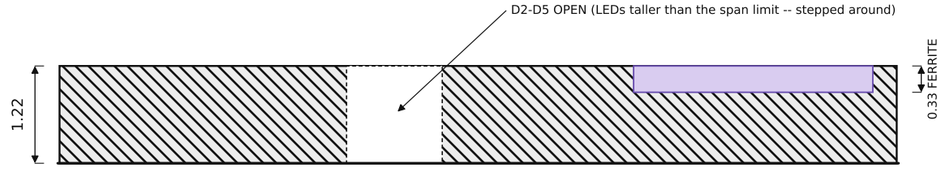
BOARD-FACING FACE SCALE 1.7:1





FLAT BOTTOM = SANDING DATUM (lap to the height fit; all pockets are on the TOP face)

SECTION B-B (typ., across brace) SCALE 17:1

bottom faces the shell floor • top faces the board back

NOTES

1. PART = SOLAR-GLOW DRH DIFFUSER BRACE [LITE]. A DROP-IN INSERT, NOT A FASTENED OR BONDED PART. 3D STEP GOVERNS ALL GEOMETRY; THE STL IS THE PRINT SOURCE.
2. PROCESS: SLA / RESIN 3D PRINT (e.g. PCBWay). MATERIAL: TOUGH WHITE RESIN. IT MUST BE OPAQUE-WHITE AND NON-CONDUCTIVE - NO CARBON / GRAPHITE-FILLED RESIN (WEAKLY CONDUCTIVE): THE BRACE RESTS ON GND / VS / SIGNAL COPPER, SO A DIELECTRIC IS REQUIRED. WHITE ALSO DRIVES THE WINDOW BACKING (NOTE 6).
3. FIT: PRINT ~0.1 PROUD IN HEIGHT AND SAND THE FLAT BOTTOM (DATUM) DOWN TO A ZERO-AIR FIT IN THE 1.22 CAVITY. ALL POCKETS ARE ON THE TOP FACE, SO THE BOTTOM LAPS FLAT ON GLASS WITHOUT TOUCHING THEM. DO NOT SAND THE TOP (IT SETS THE POCKET DEPTHS).
4. PRECISION FIT, NO RATTLE: THE OUTER EDGES CONTACT THE FLAT CAVITY WALLS AT ~0.05 WHERE THE FOOTPRINT REACHES THEM (COMPUTED EXTENT x 2.55-49.50, y 2.05-86.85; E EDGE STEPPED -- THE PLAN IS THE COMPUTED FOOTPRINT). THE FOUR CORNER BOSSES (r2.6) + ROUNDED CORNERS ARE RELIEVED (NEED NOT FIT). THE FOOTPRINT RUNS S->N WALL BUT PUTS NO RESIN BEHIND ANY OF THE 4 PANEL SOLDER TABS (THE NEAREST IS 14.1 FROM A TAB CENTRE): THE 4 PIECES DROPPED TO KEEP THE BRACE ONE PART (202 mm2) WOULD BACK 4 OF THEM; POCKETS KEY TO THE BOARD.
5. FERRITE CHANNEL (OVER THE NFC COIL): OPEN-ENDED, WALLED 12.40 = FERRITE 12.0 + 0.20 EACH SIDE (CRITICAL - EDGE-LIMITED), OPEN AT BOTH Y-ENDS, 0.33 DEEP. FERRITE (Wurth WE-FSFS 364006, NOMINAL 12 × 26 mm, EVEN ON THE 2mm SCORE GRID) IS PSA'd IN; LENGTH IS FORGIVING AND MAY OVERHANG THE ENDS SLIGHTLY.
6. WINDOW BACKING: SOLID WHITE RESIN FILLS THE MONOGRAM-WINDOW FOOTPRINT BEHIND THE FR4, MINUS OPEN D2-D5 CUTOUTS (THE LEDs EXCEED THE SPAN LIMIT: STEPPED AROUND). NO APERTURE, NO FLOOR TAPE. THE POCKET CLEARANCE DOUBLES AS A RESERVOIR IF A VISCOUS OPTICAL GEL IS PRE-FILLED AT FINAL ASSEMBLY (OPTIONAL).
7. REMOVABLE / NOT BONDED: THE BRACE MUST LIFT OUT FOR NFC C9 TRIM DURING BENCH BRING-UP. KEEP IT DRY-FIT WHILE ITERATING; ADD ANY GEL ONLY ON THE FINAL CARD.
8. ALL COMPONENT POCKETS ARE BLIND. PARTS TALLER THAN 0.60 CANNOT BE COVERED AT ALL (WEB WOULD FALL BELOW 0.40), SO THE FOOTPRINT STEPS AROUND THEM: SC1, SW2, C13, U8, D4, U1, U7, D5, C27, C25, C26, C9, C4, SC3, FB1, D3, L2, D2, U3, C22, SC4, Q2, SC2, C23. U7 (THE 0.90 DFN-8) IS TALLER THAN THE 0.60 SPAN LIMIT HERE, SO IT IS STEPPED AROUND, NOT POCKETED.

SOLAR-GLOW DRH — RESIN DIFFUSER BRACE [LITE]	
DWG solar-glow-drh-diffuser-brace-lite	REV B
MATERIAL TOUGH WHITE SLA (opaque, non-conductive)	
UNITS mm	SCALE AS NOTED
ENVELOPE 47.0 x 84.8 x 1.22 process: SLA print	SHEET 1/1